

CURRENCY CONVERTER

API Documentation

FUTM-SWE-221 Project 6 Developed by Aisha
API: Open Exchange Rates openexchangerates.org

Project Overview

This is Project 6 of the FUTM-SWE-221 Individual Software Development Project assignment. The application is a Currency Converter that integrates the Open Exchange Rates REST API to fetch live exchange rates for five currencies: NGN (Nigerian Naira), USD (US Dollar), EUR (Euro), GBP (British Pound), and GHS (Ghanaian Cedi). The free plan always returns rates relative to USD, so the conversion formula applied is: converted = amount multiplied by (rate_TO divided by rate_FROM).

SDLC Phases

Planning	Identified five core currencies relevant to Nigeria. Selected Open Exchange Rates as the API provider because it is free, requires no card, and supports over 170 currencies.
Analysis	End user identified as a market trader persona who needs quick and accurate NGN conversions on a mobile device without signing up for anything.
Design	Designed a single-page mobile-first converter with a clean layout, large result display, and a swap button so users can reverse currency pairs instantly.
Implementation	Built with PHP and cURL. Calls the /latest.json endpoint and applies the formula: amount x (rate_TO / rate_FROM) to handle all currency pairs including non-USD pairs.
Testing	Tested conversions across all five currency combinations including NGN to USD, USD to NGN, NGN to GHS, and EUR to GBP. Also tested edge cases such as zero amounts and identical currency pairs.
Deployment	Deployed as a single PHP file that runs on any web server with cURL enabled. No database or external framework is required.

API Information

Provider	Open Exchange Rates
Base URL	https://openexchangerates.org/api
App ID	9069645b9f1f4dc4bb634627a5e6d9d4
Authentication	Query parameter: app_id=YOUR_APP_ID
Response Format	JSON
Free Plan Quota	1,000 requests per month. Rates update every hour.
Base Currency	Always USD on the free plan. Custom base requires a paid plan.
Conversion Formula	converted = amount x (rate_TO / rate_FROM)
Documentation	https://docs.openexchangerates.org

Test Currencies

Code	Currency	Symbol	Rate (per USD)	Role
NGN	Nigerian Naira	₦	1370.500000	Base / Anchor
USD	US Dollar	\$	1.000000	Target
EUR	Euro	€	0.880038	Target

GBP

British Pound

0.757958

Target

Ghanaian Cedi

GH μ

11.225000

Target

API Requests and Responses

All five requests below were sent using the Open Exchange Rates API. The request URL, parameters, and the full JSON body returned by the server are shown for each test.

1. Test 1: Check API Usage and Quota

This request checks how many of the 1,000 monthly free requests have been used and confirms the App ID is active.

Method: GET
URL: https://openexchangerates.org/api/usage.json?app_id=9069645b9f1f4dc4bb634627a5e6d9d4
Param: app_id=9069645b9f1f4dc4bb634627a5e6d9d4 (Required: Yes)
Status: 200 OK

Response Body:

```
{
  "status": 200,
  "data": {
    "app_id": "9069645b9f1f4dc4bb634627a5e6d9d4",
    "status": "active",
    "plan": {
      "name": "Free",
      "quota": "1000 requests / month",
      "update_frequency": "3600s"
    },
    "usage": {
      "requests": 0,
      "requests_quota": 1000,
      "requests_remaining": 1000,
      "days_elapsed": 0,
      "days_remaining": 30,
      "daily_average": 0
    }
  }
}
```

2. Test 2: Get Currency Names List

Retrieves the human-readable names of all 170+ supported currencies. This is a public endpoint and does not require an App ID.

Method: GET
URL: <https://openexchangerates.org/api/currencies.json>
Param: none = Public endpoint. No authentication required. (Required: No)
Status: 200 OK

Response Body:

```
{
  "AED": "UAE Dirham",
  "EUR": "Euro",
  "GBP": "Pound Sterling",
  "GHS": "Ghanaian Cedi",
  "NGN": "Nigerian Naira",
  "USD": "United States Dollar"
}
```

3. Test 3: Get Latest Rates for All 5 Currencies

Fetches live exchange rates for all five test currencies. The base is always USD on the free plan. Rates are updated every hour.

```
Method:      GET
URL:         https://openexchangerates.org/api/latest.json?
app_id=9069645b9f1f4dc4bb634627a506d9d4&symbols=NGN,USD,EUR,GBP,GHS
Param:       app_id = 9069645b9f1f4dc4bb634627a506d9d4 (Required: Yes)
Param:       symbols = NGN,USD,EUR,GBP,GHS (Required: No)
Status:      200 OK
```

Response Body:

```
{
  "disclaimer": "Usage subject to terms: https://openexchangerates.org/terms",
  "license": "https://openexchangerates.org/license",
  "timestamp": 1782277200,
  "base": "USD",
  "rates": {
    "EUR": 0.880038,
    "GBP": 0.757958,
    "GHS": 11.225,
    "NGN": 1370.5,
    "USD": 1
  }
}
```

4. Test 4: Convert 10,000 NGN to USD

Converts 10,000 Nigerian Naira to US Dollars. Since the free plan base is always USD, the formula used is: converted = amount x (rate_TO / rate_FROM). Result: 10000 x (1 / 1370.5) = 7.2969 USD.

```
Method:      GET
URL:         https://openexchangerates.org/api/latest.json?
app_id=9069645b9f1f4dc4bb634627a506d9d4&symbols=NGN,USD
Param:       app_id = 9069645b9f1f4dc4bb634627a506d9d4 (Required: Yes)
Param:       symbols = NGN,USD (Required: No)
Status:      200 OK
```

Response Body:

```
{
  "disclaimer": "Usage subject to terms: https://openexchangerates.org/terms",
  "license": "https://openexchangerates.org/license",
  "timestamp": 1782277200,
  "base": "USD",
  "rates": {
    "NGN": 1370.5,
    "USD": 1
  }
}
```

Calculation applied on the client side:

```
amount      = 10,000 NGN
rate_FROM    = 1370.5 (NGN per 1 USD)
rate_TO      = 1      (USD per 1 USD)
converted    = 10000 x (1 / 1370.5)
result       = 7.2969 USD
```

5. Test 5: Convert 1 USD to NGN (Reverse Pair)

Converts 1 US Dollar back to Nigerian Naira, testing the reverse currency pair. Formula: converted = amount x (rate_TO / rate_FROM). Result: 1 x (1370.5 / 1) = 1,370.50 NGN.

```
Method:    GET
URL:       https://openexchangerates.org/api/latest.json?
app_id=9069645b9f1f4dc4bb634627a5e6d9d4&symbols=NGN,USD
Param:     app_id=9069645b9f1f4dc4bb634627a5e6d9d4 (Required: Yes)
Param:     symbols = NGN,USD (Required: No)
Status:    200 OK
```

Response Body:

```
{
  "disclaimer": "Usage subject to terms: https://openexchangerates.org/terms",
  "license": "https://openexchangerates.org/license",
  "timestamp": 1782277200,
  "base": "USD",
  "rates": {
    "NGN": 1370.5,
    "USD": 1
  }
}
```

Calculation applied on the client side:

```
amount      = 1 USD
rate_FROM   = 1      (USD per 1 USD)
rate_TO     = 1370.5 (NGN per 1 USD)
converted    = 1 x (1370.5 / 1)
result      = 1,370.50 NGN
```